Chapter 11: Projects

Project 1: Simple Calculator

In this project, we'll build a simple console-based calculator that performs basic arithmetic operations. This project will help you apply your knowledge of Python, including variables, user input, functions, and control flow. By the end of this project, you'll have a solid understanding of how to create a simple program that can handle user input, process it, and return a result.

Objectives:

- Understand how to take input from the user.
- Implement basic arithmetic operations (addition, subtraction, multiplication, and division).
- Use functions to organize code logically.
- Implement error handling for common issues like division by zero and invalid input.
- Utilize loops to allow for multiple calculations without restarting the program.

Step-by-Step Guide to Build the Simple Calculator

Step 1: Define the Functions for Basic Operations

Start by defining the functions that will handle the basic arithmetic operations. These will be simple functions that take two parameters (the numbers to operate on) and return the result of the operation.